LAB 4: SUBNETTING AND SUPERNETTING USING CISCO PACKET TRACER

OBJECTIVES

1. To understand the logical division of IP networks through Subnetting and the aggregation of networks through Supernetting.

2. To design a variable length or fixed length subnet mask scheme and implement the topology using Cisco Packet Tracer to visualize the flow of packets between different subnets via a Router.

SOFTWARE AND HARDWARE REQUIREMENTS

1. Cisco Packet Tracer (Version 6.2 or higher)

2. Windows PC/Laptop

THEORY

1. **Subnetting**

Subnetting is the process of breaking one large physical network into several smaller logical networks called subnets. This is done by borrowing bits from the host part of an IP address to create additional network identifiers.

How it works: A subnet mask is used to extend the network portion of the IP address. For example, borrowing 2 bits from a Class C (/24) network results in 4 smaller subnets. Routers use the subnet mask to identify which subnet an IP address belongs to and route traffic correctly.

Why it's used: Subnetting improves network performance, management, and security by reducing broadcast traffic. It keeps local traffic within the same subnet and prevents unnecessary data from spreading across the entire network, making the network more efficient and organized.

2. **Supernetting**

Supernetting, also known as **Classless Inter-Domain Routing (CIDR)** or route summarization, is the reverse of subnetting. It allows multiple smaller, consecutive networks to be merged into a single larger network.

How it works: Supernetting is done by reducing the number of network bits in the subnet mask, effectively shifting the network boundary to the left. This creates one summary (aggregate) address that represents a group of contiguous networks instead of listing each one separately.

Why it's used: Supernetting is mainly used to reduce the size of routing tables in routers. Smaller routing tables require less memory and processing power, which speeds up routing decisions and improves overall Internet efficiency and scalability

NETWORK DESIGN

Subnetting

Calculations

Base network: 192.168.1.0/24

Required number of subnets: 4

Number of IP address per subnet: 64 (Block size)

/26 (Borrowed 2 bits: 2^2 = 4 subnets)

Subnet Mask: 255.255.255.192

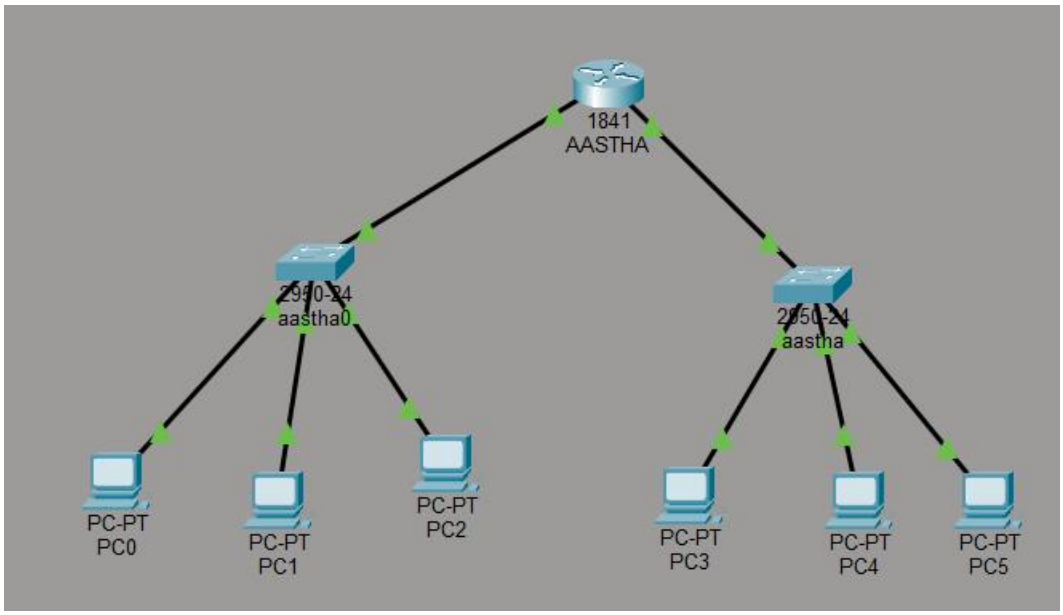
Subnet Calculation Table

Subnet	Network ID	1st Host	Last Host	Broadcast ID
1	192.168.1.0	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193	192.168.1.254	192.168.1.255

Network Topology

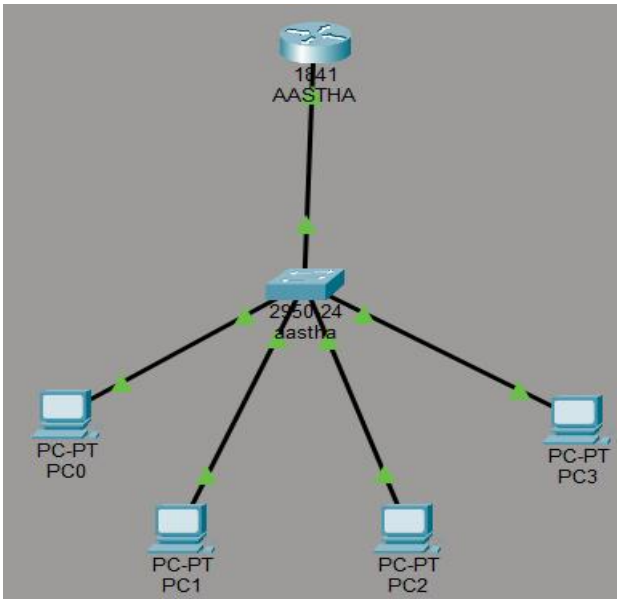
Subnetting:

A topology was created using Router0 to connect two subnets: 192.168.1.0/26 and 192.168.1.64/26. Switch0 connects PC0, PC1, and PC2 in the first subnet, while Switch1 connects PC3, PC4, and PC5 in the second subnet. Both switches are connected to the router interfaces.



Supernetting:

The topology consists of one router, one switch, and four PCs (PC0–PC3). All PCs connect to the switch, which is connected to the router.



Configuration table

Subnetting

Device	IPV4	Subnetmask	Default gateway
Router0 (FastEthernet0/0)	192.168.1.1	255.255.255.192	N/A
Router0 (FastEthernet0/1)	192.168.1.65	255.255.255.192	N/A
PC0 (Subnet 1)	192.168.1.2	255.255.255.192	192.168.1.1
PC1 (Subnet 1)	192.168.1.3	255.255.255.192	192.168.1.1
PC2 (Subnet 1)	192.168.1.4	255.255.255.192	192.168.1.1
PC3 (Subnet 2)	192.168.1.66	255.255.255.192	192.168.1.65
PC4 (Subnet 2)	192.168.1.67	255.255.255.192	192.168.1.65
PC5 (Subnet 2)	192.168.1.68	255.255.255.192	192.168.1.65

Supernetting

Device	IPV4	Subnetmask	Default gateway
Router0 (FastEthernet0/0)	192.168.0.1	255.255.252.0	N/A
PC0	192.168.0.10	255.255.252.0	192.168.0.1
PC1	192.168.1.10	255.255.252.0	192.168.0.1
PC2	192.168.2.10	255.255.252.0	192.168.0.1
PC3	192.168.3.10	255.255.252.0	192.168.0.1

For subnetting:

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time=4ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=12ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>
```

Ping PC0 to PC4

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.67

Pinging 192.168.1.67 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.67: bytes=32 time<1ms TTL=127
Reply from 192.168.1.67: bytes=32 time=1ms TTL=127
Reply from 192.168.1.67: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.1.67:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping PC0 to PC3

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Result

The network was successfully configured and tested for connectivity. Ping requests from **PC0 (192.168.1.2)** to **PC2 (192.168.1.4)** within Subnet 1 and from **PC0** to **PC4 (192.168.1.67)** in Subnet 2 were completed successfully. Similar ping tests were also performed for the supernetting configuration, and all packets were transmitted and received correctly, confirming that the network connections were properly set up.

Discussion and conclusion

During the lab session, we implemented the concepts of **subnetting and supernetting** to gain a better understanding of IP address management and efficient network design. Subnetting allowed us to divide a large network into multiple smaller subnets, which helped in better utilization of IP addresses and reduced unnecessary network traffic. It also improved network performance by limiting the broadcast domain and making the network easier to manage.

On the other hand, supernetting enabled the combination of multiple smaller networks into a single larger network. This simplified routing by reducing the number of routing entries required in routing tables. Through these practical exercises, we learned how subnetting and supernetting play a crucial role in optimizing network performance, improving scalability, and ensuring effective communication in modern networks.

Hence, the lab was successfully completed with proper understanding and practical implementation of subnetting and supernetting using **Cisco Packet Tracer**.